

Introduction to Stochastic Processes

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Stochastic processes

A **stochastic process**, or **random process**, is a mathematical model representing a magnitude that changes over time so that a random variable is associated with each specific moment in time.

Definition

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $R_t \subseteq \mathbb{R}$. A stochastic process X is defined as a mapping

$$\begin{aligned} X : R_t \times \Omega &\longrightarrow \mathbb{R} \\ (t, \omega) &\mapsto X(t, \omega) \end{aligned}$$

such that $X(t, \cdot) : \Omega \rightarrow \mathbb{R}$ is a random variable for each $t \in R_t$.

- ▶ Let $t \in R_t$. Since $X(t, \cdot)$ is a random variable, for all $x \in \mathbb{R}$, the subset of outcomes

$$\{\omega \in \Omega: X(t, \omega) \leq x\}$$

is an event (i.e., an element of the σ -algebra $\mathcal{F}$).

- ▶ In many applications the parameter t means **time**.
- ▶ The dependence on ω is usually omitted and one writes X or $X(t)$ to denote the stochastic process.

Stochastic processes

Therefore,

- ▶ A stochastic process X is a collection of random variables indexed by the parameter t ,

$$\{X(t, \omega) \equiv X_t(\omega): t \in R_t\}$$

- ▶ X is also a collection of functions of time,

$$\{X(t, \omega) \equiv X_\omega(t): \omega \in \Omega\}$$

The function $X_\omega(t)$ is the **realization** (or trajectory) of the process associated to the outcome $\omega \in \Omega$.

Discrete-time processes

If the set R_t is finite or denumerable, we say that X is a **discrete-time** process. In particular,

- ▶ If $R_t = \{t_1, t_2, \dots, t_n\}$ is a finite set, then

$$X \equiv (X(t_1), X(t_2), \dots, X(t_n))$$

is a random vector.

- ▶ If $R_t = \{t_1, t_2, \dots, t_n, \dots\}$ is denumerable, then

$$X[n] \equiv \{X(t_n) : n \geq 1\}$$

is a sequence of random variables.

Continuous-time processes

When the set R_t is uncountable, one says that X is a **continuous-time** process.

Usual cases are:

$$X(t), \quad t \in [a, b]$$

$$X(t), \quad t \in (0, \infty)$$

$$X(t), \quad t \in \mathbb{R}$$

State space

According to the values taken by the process (states) we have:

- ▶ $X(t)$ is a **discrete-state** process if the image set of the mapping X is finite or denumerable. So, for each $t \in R_t$, one has a discrete random variable.
- ▶ Otherwise, $X(t)$ is a **continuous-state** process. Usually, $X(t)$ is a continuous random variable for all $t \in R_t$.

Distribution and density functions

One can consider, for each $t \in R_t$, the probability distribution function of the random variable $X(t)$:

$$F_X(x; t) \equiv F_{X(t)}(x) = \mathbb{P}(X(t) \leq x)$$

- ▶ $F_X(x; t)$ is called the **first-order distribution function** of the process.
- ▶ $F_X(x; t)$ describes the time evolution of the probability law of the collection of random variables $\{X(t): t \in R_t\}$.

Distribution and density functions

For a continuous-state stochastic process, $X(t)$ is a continuous random variable for any $t \in R_t$. So we can consider the **first-order density**:

$$f(x; t) \Delta x \approx \mathbb{P}\{x \leq X(t) \leq x + \Delta x\}$$

More formally,

$$f(x; t) = \lim_{\Delta x \rightarrow 0} \frac{\mathbb{P}\{x \leq X(t) \leq x + \Delta x\}}{\Delta x}$$

- Notice that $f_X(x; t)$ is, for each $t \in R_t$, the probability density function $f_{X(t)}(x)$ of the continuous variable $X(t)$.

Distribution and density functions

The known relations between distribution and density functions hold. For instance,

$$f_{X(t)}(x) = F'_{X(t)}(x),$$

that is to say,

$$f_X(x; t) = \frac{\partial F_X(x; t)}{\partial x}$$

Analogously,

$$F_{X(t)}(x) = \int_{-\infty}^x f_{X(t)}(u) \, du,$$

that is,

$$F_X(x; t) = \int_{-\infty}^x f_X(u; t) \, du$$

Distribution and density functions

Similarly, if $X(t)$ is a discrete-state process that takes the values $x_1, \dots, x_k, \dots$, we can define the **first-order probability function** as follows:

$$p_X(x_k; t) \equiv p_{X(t)}(x_k) = \mathbb{P}(X(t) = x_k), \quad k \geq 1$$

So we have

$$F_X(x; t) = \sum_{x_k \leq x} p_X(x_k; t).$$

Distribution and density functions

Definition

The n -order distribution function of the stochastic process $X(t)$, denoted as $F_X(x_1, \dots, x_n; t_1, \dots, t_n)$, represents the joint distribution function of n random variables $X(t_1), \dots, X(t_n)$ arising from $X(t)$ at times $t_1, \dots, t_n \in R_t$. Thus, it can be expressed as:

$$\begin{aligned} F_X(x_1, \dots, x_n; t_1, \dots, t_n) &\equiv F_{X(t_1), \dots, X(t_n)}(x_1, \dots, x_n) \\ &= \mathbb{P}(X(t_1) \leq x_1, \dots, X(t_n) \leq x_n) \end{aligned}$$

- ▶ A complete characterisation of the stochastic process $X(t)$ requires the knowledge of $F_X(x_1, \dots, x_n; t_1, \dots, t_n)$ for all possible choices of $t_1, \dots, t_n$ and for all $n \geq 1$.

Distribution and density functions

For a continuous-state process, the n -order density function is the joint density of the random variables $X(t_1), \dots, X(t_n)$:

$$\begin{aligned} f_X(x_1, \dots, x_n; t_1, \dots, t_n) &\equiv f_{X(t_1), \dots, X(t_n)}(x_1, \dots, x_n) \\ &= \frac{\partial^n F_X(x_1, \dots, x_n; t_1, \dots, t_n)}{\partial x_1 \cdots \partial x_n} \end{aligned}$$

Analogously, if $X(t)$ is a discrete-state process, the n -order probability function is

$$p_X(x_{k_1}, \dots, x_{k_n}; t_1, \dots, t_n) = \mathbb{P}(X(t_1) = x_{k_1}, \dots, X(t_n) = x_{k_n})$$

Distribution and density functions

The usual relations and properties of these functions are maintained. For example,

$$\int_{-\infty}^{\infty} f_X(x; t) dx = 1$$

$$\int_{-\infty}^{\infty} f_X(x_1, x_2; t_1, t_2) dx_1 = f_X(x_2; t_2)$$

$$\lim_{x_2 \rightarrow \infty} F_X(x_1, x_2; t_1, t_2) = F_X(x_1; t_1)$$

$$p_X(x_{k_1}; t_1) = \sum_{k_2} p_X(x_{k_1}, x_{k_2}; t_1, t_2)$$

Mean and autocorrelation

Definition

The mean function of the process is $m_X(t) = \mathbb{E}(X(t))$

- ▶ For each $t \in R_t$, $m_X(t)$ is the expected value of the random variable $X(t)$.
- ▶ Therefore, for a continuous-state process,

$$m_X(t) = \int_{-\infty}^{\infty} x f_X(x; t) dx$$

- ▶ Analogously, for a discrete-state process,

$$m_X(t) = \sum_k x_k p_X(x; t)$$

Mean and autocorrelation

For discrete-time processes, the these formulas are as follows:

$$m_X[n] = \mathbb{E}(X[n])$$
$$= \begin{cases} \int_{-\infty}^{\infty} x f_X(x; n) dx, & \text{continuous-state} \\ \sum_k x_k p_X(x_k; n), & \text{discrete-state} \end{cases}$$

Mean and autocorrelation

Definition

The autocorrelation function of the stochastic process $X(t)$ is defined as

$$R_X(t_1, t_2) = \mathbb{E}(X(t_1)X(t_2)).$$

- ▶ $R_X(t_1, t_2)$ is the joint moment m_{11} of the random variables $X(t_1)$ and $X(t_2)$.
- ▶ It is a symmetric function, $R_X(t_1, t_2) = R_X(t_2, t_1)$.
- ▶ If $t_1 = t_2 = t$, we have $R_X(t, t) = \mathbb{E}(X^2(t))$. This function is called the **average power** of $X(t)$.

Mean and autocorrelation

- ▶ If $X(t)$ is a continuous-state process,

$$R_X(t_1, t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_X(x_1, x_2; t_1, t_2) dx_1 dx_2,$$

- ▶ Otherwise, if $X(t)$ a discrete-state process we have that

$$R_X(t_1, t_2) = \sum_{k_1} \sum_{k_2} x_{k_1} x_{k_2} p_X(x_{k_1}, x_{k_2}; t_1, t_2).$$

- ▶ For discrete-time processes we can write analogous expressions.

Autocovariance

Definition

The autocovariance function of the stochastic process $X(t)$ is defined as

$$C_X(t_1, t_2) = \mathbb{E}((X(t_1) - m_X(t_1))(X(t_2) - m_X(t_2))).$$

- ▶ Expanding this expression,

$$C_X(t_1, t_2) = R_X(t_1, t_2) - m_X(t_1)m_X(t_2)$$

- ▶ Notice that $C_X(t, t) = \mathbb{E}((X(t) - m_X(t))^2)$ is the variance of the random variable $X(t)$.

Stationary processes

A stochastic process $X(t)$ is defined as **strict-sense stationary** if its statistical properties remain unchanged under shifts of the initial time $t = 0$.

Specifically, the processes $X(t)$ and $X(t + c)$ exhibit identical statistical characteristics for all values of c such that $t + c$ remains within the time-frame R_t .

Definition

The stochastic process $X(t)$ is strict-sense stationary if, for all $n \geq 1$, for all $t_1, \dots, t_n$, and for all c such that $t + c \in R_t$, the collections of random variables $\{X(t_1), \dots, X(t_n)\}$ and $\{X(t_1 + c), \dots, X(t_n + c)\}$ have the same joint probability distribution.

Stationary processes

Therefore, for all $n \geq 1$, for all $t_1, t_2, \dots, t_n$, and for all c , we must have

$$F_X(x_1, \dots, x_n; t_1, \dots, t_n) = F_X(x_1, \dots, x_n; t_1 + c, \dots, t_n + c)$$

- ▶ When this condition holds for $n = 1, 2, \dots, k$, we say that $X(t)$ is **k -order stationary**.
- ▶ Hence, a strict-sense stationary process is k -order stationary for all $k \geq 1$.

Stationary processes

In particular, taking $n = 1$ we deduce that

$$F_X(x; t) = F_X(x; t + c) \quad \text{for all } c$$

That is to say, the first-order distribution function of a stationary process is time-invariant:

$$F_X(x; t) \equiv F_X(x)$$

Analogously, if $X(t)$ is a continuous-state process, then

$$f_X(x; t) \equiv f_X(x)$$

Stationary processes

Theorem

The mean function of a stationary stochastic process is constant.

Proof:

$$m_X(t) = \int_{-\infty}^{\infty} x f_X(x; t) dx = \int_{-\infty}^{\infty} x f_X(x) dx \equiv m_X.$$

- More generally, any property concerning a single random variable from the process will remain time-invariant.

Stationary processes

For $n = 2$ we have

$$F_X(x_1, x_2; t_1, t_2) = F_X(x_1, x_2; t_1 + c, t_2 + c), \text{ for all } c$$

So, taking $c = -t_1$, we deduce that

$$F_X(x_1, x_2; t_1, t_2) = F_X(x_1, x_2; 0, t_2 - t_1),$$

and, hence, if $\tau = t_2 - t_1$, then

$$F_X(x_1, x_2; t_1, t_2) \equiv F_X(x_1, x_2; \tau)$$

- If $X(t)$ is stationary, the joint distribution of two random variables $X(t_1)$ and $X(t_2)$ (arising from the process) only depends on time through the lag $\tau = t_2 - t_1$.

Theorem

The autocorrelation function of a k -order stationary process $X(t)$ ($k \geq 2$) only depends on the lag τ :

$$R_X(\tau) = \mathbb{E}(X(t)X(t + \tau)),$$

where this expectation is independent of t .

In particular (taking $t = 0$),

$$R_X(\tau) = \mathbb{E}(X(0)X(\tau)).$$

Properties of $R_X(\tau)$

- ▶ $R_X(\tau)$ is an even function:

$$R_X(-\tau) = \mathbb{E}(X(t)X(t - \tau)) = \mathbb{E}(X(t' + \tau)X(t')) = R_X(\tau)$$

- ▶ The average power of $X(t)$ is constant:

$$\mathbb{E}(X^2(t)) = R_X(0) \geq 0$$

- ▶ We have that $|R_X(\tau)| \leq R_X(0)$ for any τ .

Indeed,

$$0 \leq \mathbb{E}((X(0) \pm X(\tau))^2) = 2R_X(0) \pm 2R_X(\tau)$$

Wide-sense stationary processes

Definition

A stochastic process $X(t)$ is wide-sense stationary if the following conditions hold:

- ▶ The mean function is time-invariant:

$$m_X(t) = m_X.$$

- ▶ The autocorrelation depends only on τ :

$$R_X(t, t + \tau) = \mathbb{E}(X(t)X(t + \tau)) = R_X(\tau).$$

Wide-sense stationary processes

For discrete-time processes, the previous definition can be expressed as follows:

Definition

A random sequence $X[n]$ is wide-sense stationary if

- ▶ The mean is constant, that is $m_X[n] = m_X[0]$ for all n .
- ▶ The autocorrelation $R_X[n, n+k] \equiv R[k]$ depends only on k .